Speed of light

The speed of light in free space is an important and intriguing constant of nature. Whether the light comes from a laser on a desktop or from a distant galaxy that is hurtling away at fantastic speeds, the measurement of the speed of light will yield the same constant value. In more precise terminology, the speed of light is independent of the relative velocities of the light source and the observer.

As Einstein first presented in his special theory of relativity, the speed of light is critically important in some surprising ways:

1. The speed of light establishes an upper limit to the speed that may be imparted to any material object.

2. Objects moving near the speed of light follow a set of physical laws drastically different, not only from Newton's Laws, but from the basic assumptions of human intuition.

It is not surprising that a great deal of time and effort has been invested in measuring the speed of light. Some of the most accurate measurements were made by Albert Michelson between 1926 and 1929. Michelson measured the speed of light in air to be 2.99712×10^8 m/s. From this result he deduced the speed of light in free space to be 2.99796×10^8 m/s. Today, the speed of light in free space (i.e., a vacuum) is, by definition, 2.997924562×10^8 m/s exactly.

Our approach to the measurement of the speed of light will be direct and intuitive, and simpler then Michelson's experiment. We will use a diode laser as our light source. We will send pulses of light across the lab, measure how long they take to come back, and deduce the speed of light from the knowledge that light travels at constant speed.

Laser light safety notes

We will use a relatively safe, low power laser. Nevertheless, take the following precautions:

1. Never look directly into the laser beam, either directly, or as it is reflected from a mirror or other surface.

2. Keep the optical path at floor level and keep your eyes well above floor level at all times.

3. Only people involved with the experiment should be nearby.

Experimental setup

1. Place the breadboard with the laser and photodiode detector on the floor. There should be at least 8-12 meters of space in front of the laser.

2. The laser diode has two connections: the power supply that connects to an outlet and the trigger signal that is sent to the oscilloscope. The trigger signal cable should have a 50-Ohm at the oscilloscope.

3. Connect the photodiode output to channel 2 on the oscilloscope.

4. Align the laser so that it propagates along a line on the floor and that its height is constant throughout. It will make alignment easier afterwards. Once this alignment is done,

5. Place the mirror approximately 1 m in front of the laser and use it to reflect the beam on the light receiver sensor.

6. Check the signal on the oscilloscope. You should trigger on channel 1. On channel 2, you should see a strong (100s of mV) that is stable in amplitude and time. If you do not see the signal clearly, make sure the returning laser beam is well-centered on the light receiver sensor.

Procedure

- Using a tape measure, record the total path length of the light pulses.
- Using the cursors on the oscilloscope, measure the time delay between the two traces. One of the cursors should be set to a fixed feature on the triggering signal (channel 1). The other cursor should be set to a consistent feature near the onset of the photodiode signal.
- Determine the uncertainty in your time delay measurement and your distance measurement; there are multiple ways to do this, but be sure to justify the methods you choose. *It is a good idea to try different techniques for your measurements until you find a method that will be as precise as possible.*
- Once you have finalized your method of measuring the arrival time of the signal, move the mirror back about 50 cm and repeat the measurement. Continue making measurements over as large of a range of distances as you are able to determine reliably. Note that *your uncertainties in your position and time delay may change as you change distances*; again, be sure to justify the uncertainties you determine for your measurements.

Analysis

1. Make a graph of the time delay as a function of round trip distance. Find the slope of this graph and its uncertainty. Convert this slope (and uncertainty) into a value for the speed of light (and uncertainty).
2. Compare your result to the speed of light in a vacuum quantitatively.
3. Your experiment was not performed in a vacuum, but rather in air. How is this expected to affect your measurement? Use your measurements to determine an estimate for the index of refraction of the air in the room and your uncertainty on this index of refraction. Discuss whether your experiment is able to detect the effect of the air with significance or not.
4. What do you expect are the dominant sources of random and systematic uncertainty in your measurement? How large of an effect do these have on your measurement, and how could you reduce them?